P1355R2
Exposing a narrow contract for ceil2

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http://wg21.link/P1355R2

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Audience:
LEWG, LWG, SG6

Project:
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Abstract

ceil2 promises an unspecified value for out-of-bounds arguments. Out-of-bounds arguments should instead be undefined behavior.

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§ 1. Background

During the review of [P0556R1], LEWG requested `ceil2` return an unspecified value for out-of-bounds arguments. [P0556R3] was adopted in Rapperswil with this specification. [N4791] ([bit.pow.two] 25.5.4 "Integral powers of 2"):

> Returns: The minimal value `y` such that `ispow2(y)` is `true` and `y >= x`; if `y` is not representable as a value of type `T`, the result is an unspecified value.

At the [SAN] meeting, LEWG took a poll reconsidering this decision at the request of SG6:

> Change ceil2 to hard UB, ill-formed in constexpr
>
> | SF | F | N | A | SA |
> |---|---|---|---|---|
> | 11 | 5 | 1 | 0 | 0 |

As discussed in [P1233R0] (out-of-bounds `shift_left` / `shift_right`), silently accepting out-of-bounds values can hide bugs. Making this explicitly undefined behavior allows it to be detected and flagged by analysis tools.

§ 2. Proposal

Wording relative to [N4791]:

```
template<class T>
  constexpr T ceil2(T x) ~~noexcept~~;
```

- Let *N* be the smallest power of 2 greater than or equal to `x`.
- *Constraints*: ~~*Remarks*: This function shall not participate in overload resolution unless~~ `T` is an unsigned integer type ([basic.fundamental]).
- *Expects*: *N* is representable as a value of type `T`.
- *Returns*: *N* ~~The minimal value `y` such that `ispow2(y)` is `true` and `y >= x`; if `y` is not representable as a value of type `T`, the result is an unspecified value.~~
- *Throws*: Nothing.
- *Remarks*: A function call expression that violates the precondition in the Expects element is not a core constant expression ([expr.const]).

§ 3. Discussion

For values of `x` that fail to satisify the *Expects* precondition, this results in undefined behavior ([res.on.required] 15.5.4.11).

In earlier drafts, `noexcept` and a narrow contract were intentional based on the feedback at the [SAN] meeting. This direction was reversed during the [KonaLEWGReview].

§ 4. History

§ 4.1. R1 → R2

Applied feedback from [KonaLEWGReview].

- Removed `noexcept` specification, leaving decision to implementers.
- Added "*Throws*: Nothing"

Applied feedback from [LWGTeleconReview].

- Wording tweaks.

§ 4.2. R0 → R1

Applied feedback from [PostSanDiegoReview].

- Added wording suggested by Casey Carter.
- Added clarification that having a narrow contract *and* `noexcept` is intentional.

§ References

§ Informative References

[KonaLEWGReview]
P1355 LEWG Minutes. 2019-02-18. URL: http://wiki.edg.com/bin/view/Wg21kona2019/P1355

[LWGTeleconReview]
2019-05-21 LWG Telecon Review. 2019-05-21. URL: http://wiki.edg.com/bin/view/Wg21cologne2019/LWGTelecom21May

[N4791]

Working Draft, Standard for Programming Language C++. 2018-12-07. URL: http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/n4791.pdf

[P0556R1]

P0556R1: Integral power-of-2 operations. 2017-03-19. URL: http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2017/p0556r1.html

[P0556R3]

P0556R3: Integral power-of-2 operations. 2018-06-06. URL: http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p0556r3.html

[P1233R0]

Shift-by-negative in shift_left and shift_right. 2018-10-02. URL: http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p1233r0.pdf

[PostSanDiegoReview]

LEWG(I) Weekly Review - P1355R0: Exposing a narrow contract for ceil2. 2019-01-09. URL: http://lists.isocpp.org/lib-ext/2019/01/9595.php

[SAN]

Meeting minutes for P0556. 2018-11-09. URL: http://wiki.edg.com/bin/view/Wg21sandiego2018/P0556